Region Segmentation based on Fluorescent Imaging

Anonymous CVPR submission

Paper ID *****

Abstract

Hospital-associated saliva deposition can seed high-risk pathogens across shared surfaces. This work studies unsupervised segmentation of saliva deposits illuminated under UV-C, where only ten color images are available and ground truth masks only cover four centralized patches per frame. We explore a hierarchy of methods: a MATLAB-based luminance/thresholding pipeline (Method A), an exploratory k-means segmentation (Method C), and our probabilistic Gaussian mixture model (Method D) that reuses k-means centers, runs ten EM initializations per image, and selects the most likely fit by pseudo-log-likelihood. The resulting GMM probability maps are binarized with morphological cleanup and evaluated on accuracy, precision, recall, and Dice; the $K{=}4$ model achieves the highest Dice while maintaining recall above 0.96. Our comparison highlights how converting hard clusters into likelihood-ranked mixtures improves recall while still recovering reasonable precision, providing a practical unsupervised pipeline for low-data airborne contamination imagery.

1. Introduction

1.1. Problem Definition

Hospital-acquired infections (HAIs) pose a recurring threat to healthcare environments, increasing patient risk, complicating workflows, and incurring higher operational costs. HAIs spread by direct contact, droplets, aerosols, and contaminated surfaces. The problem is further harmful in busy clinical areas, where equipment is moved frequently, and patients are vulnerable. Major reviews and guidelines identify environmental reservoirs, device-related contamination, and workflow pressures as the primary factors contributing to disease propagation in hospitals, suggesting that broad infection-control policies alone are insufficient. Additionally, knowing where contamination actually accumulates is crucial, allowing for focused and effective cleaning [1, 4, 5, 8, 11].

Saliva is a common target for efforts aimed at reducing HAIs because it carries many pathogens and is easily spread during talking, coughing, and routine procedures. Once saliva lands on instruments or surfaces, its appearance changes as it dries or interacts with different materials, making it difficult to spot under normal lighting. Studies using fluorescence reveal clear differences between moist and dried saliva on various surfaces, indicating that optical imaging can detect deposits that would otherwise be missed [7, 12]. Other reviews and studies that treat saliva as a diagnostic and surveillance fluid also highlight its role in pathogen carriage and transmission, reinforcing the idea that detecting saliva specifically is a useful way to reduce HAIs [3, 6, 10].

In practice, hospitals employ a layered approach to prevention, utilizing masks, face shields, and routine cleaning and sterilization. Masks and shields reduce the spread of droplets but do not eliminate surface deposition. Experiments that use fluorescent surrogates often reveal contamination patterns invisible to the naked eye, demonstrating that barriers and cleaning must work together [9]. Mask effectiveness depends on fit, material, and how long it is worn. Leaks and saturation can still let droplets escape and contaminate nearby surfaces [2]. Face shields block splashes and large droplets but do not prevent peripheral deposition, and the shield surfaces themselves can become contaminated. Therefore, they are most effective when used in conjunction with masks [2, 9]. Routine disinfection lowers surface bioburden, but actual coverage and contact time vary in real settings. Laser-induced fluorescence (LIF) can map where droplets land and persist, providing useful feedback to improve cleaning patterns and identify missed spots [12].

Computer vision can make these imaging findings actionable. Segmentation algorithms can produce binary masks that mark saliva or fluorescent surrogates in images. These masks can guide automated or semi-automated cleaning tools to focus on contaminated regions. Fluorescence imaging and high-speed visualization provide the contrast and ground truth necessary to train and validate these algorithms across various surface types and moisture states, helping bridge the gap between lab measurements and real-

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world cleaning workflows [2, 7, 9, 12].

1.2. Literature Review

Medical image segmentation is the area of medical technology that most resembles the creation of binary masks for saliva recognition. This process labels every pixel (or voxel) in an image to isolate organs, lesions, fluids, usually by difference in contrast. For contamination detection this usually boils down to a binary mask (contaminant vs. background), but the goals are the same as other clinical tasks: get accurate boundaries, be robust to different imaging conditions, and produce outputs that can drive decisions or automated actions. Standard overlap metrics such as the Dice Coefficient and Intersection over Union (IoU) are used to judge how well a model matches expert masks, called Ground Truths (GT). These metrics guide model selection and tuning [**?**].

The methods for segmentation range from image-processing techniques to deep neural networks. Basic steps, such as contrast enhancement, thresholding, edge filters (Sobel/Laplace), and morphological cleanup, are still used for boosting visibility of small or low-contrast deposits, as well as for preprocessing images before applying the model [**? ?**]. On the learning side, U-Net and its many variants are the baseline because they combine multi-scale context with precise localization. More recent work adds attention blocks or mixes CNNs with transformers to capture longer-range context and improve small-object detection [**? ?**]. There's also a growing push to use synthetic data and generative models to expand training sets, and public codebases for droplet tasks (e.g., Droplet-ML) make it easier to reproduce experiments and try augmentation strategies [**? ?**].

The main challenges for saliva/droplet segmentation are familiar in the computer vision field. Getting enough labeled masks is expensive, droplets are tiny so class imbalance is severe, and appearance changes a lot with surface type, lighting, and drying state. That means the necessity of specialized datasets, multi-scale supervision, and careful preprocessing or domain-adaptation to generalize from lab images to real clinical scenes [**? ?**]. Practically speaking, the best routes adopt a hybrid approach, using classical enhancement to make deposits visible [**?**], training robust segmentation networks (with attention to small-object handling) [**?**], and relying on synthetic or open datasets to fill gaps [**? ?**].

1.3. Contributions

In this work we deliver:

- A structured comparison of the existing MATLAB/thresholding pipeline (Method A), the exploratory k-means clustering pipeline (Method C), and the probabilistic Gaussian mixture model pipeline (Method D).
- A GMM workflow that seeds EM with k-means centers, runs ten initializations per image, selects the best solution via pseudo-log-likelihood, and thresholds the resulting probability map before morphological cleanup.
- Empirical evidence that the $K = 4$ GMM increases Dice while keeping recall above 0.96, illustrating how likelihood-aware model selection improves the unsupervised segmentation of saliva under UV-C.

2. Dataset and Methods

2.1. Dataset description

The dataset contains ten UV-C images of saliva deposits on five substrates (aluminum, steel, white wood, PVC, and wood) with both wet and dry patches. Each frame is accompanied by a binary mask that highlights four roughly centralized saliva patches. We treat the first eight images as the tuning set and reserve the final two images for blind testing; the masks only cover the salient regions, so every method must operate in an unsupervised fashion while relying heavily on color, contrast, and spatial locality.

- **Imaging:** High-contrast UV-C photography with fluorescent saliva to accentuate deposits across heterogeneous materials.
- **Ground truth:** One binary mask per image that marks four foreground regions, which guide plausibility filters and evaluation.
- **Split:** Images 1-8 serve for parameter tuning, and 9-10 are used to report unseen performance.
- **Challenge:** The small dataset prevents supervised training, so all methods rely on heuristics shaped by saliva brightness and location.

2.2. Method A: MATLAB thresholding pipeline

Method A is a lightweight MATLAB pipeline that maximizes recall through contrast enhancement and biased thresholding. We convert RGB frames to luminance using ITU-R BT.601 weights, apply CLAHE to normalize local contrast, and tune a bias for Otsu's threshold so that the resulting binary mask includes more saliva when the default threshold underestimates the foreground.

2.2.1. Pre-processing

- **Luminance conversion:** Each pixel's RGB values are aggregated as $0.299R+0.587G+0.114B$, producing the grayscale images shown in Figures 2a and 2b.
- **CLAHE:** Contrast Limited Adaptive Histogram Equalization (Figure 2c) equalizes local histograms while limiting noise amplification, making the threshold more stable.

2.2.2. Training

Otsu's threshold pulls a level that minimizes intra-class variance on the bimodal histogram (Figure 2d), but it can underestimate saliva in uneven lighting. We therefore learn a bias factor that adjusts the threshold before binarization. The bias is updated as

$$t_{\text{new}} = t_{\text{otsu}} \times (1 - \text{bias}), \tag{1}$$

and we sweep this bias on the tuning set to maximize the Dice coefficient. Plausibility filters then remove regions that are either too large (non-saliva) or too small (noise) and reward blobs near the geometric center of the annotated patches. Figure 1 from Felipe's folder summarizes the six labeled stages (a–g) that produce the cleaned mask.

Figure 2 shows the bias tuning curve that calibrates Otsu's threshold; the selected bias of -0.18 makes the mask more permissive and closer to the annotated saliva regions (Figure 3 in the draft).

2.2.3. Testing and post-processing

With the tuned bias (Figure 2), images 9 and 10 are processed without consulting the ground truth. The same area-based filters remove unrealistic regions, but without direct knowledge of the annotated centroids, we judge centrality relative to the image center. Morphological closing (erosion followed by dilation) closes small gaps (Figure 1f), and the final mask is compared to the blind ground truth for precision/recall reporting (Figure 1f).

2.3. Method B: placeholder

Method B was reserved for future experimentation and remains unimplemented. The emphasis of this draft is on Methods A, C, and the probabilistic extension described in Method D.

2.4. Method C: exploratory k-means

Method C clusters RGB pixels with k-means. Each run uses ten random starts, and we select the solution with the smallest compactness (within-cluster sum of squares). The centroid with the highest luminance $0.299R+0.587G+0.114B$ is labeled as saliva, and the corresponding pixels form a binary mask that is cleaned with morphological opening and closing.

2.4.1. Pre-processing

No pre-processing beyond using raw RGB pixels was required—the UV-C contrast is already sufficient to separate saliva from the background, and Gaussian or median filters degraded performance.

2.4.2. Clustering

- Ten runs per image keep the best result by compactness.
- The brightest center identifies saliva, and its assigned pixels become the foreground mask.
- Each mask is cleaned with morphological operations (opening and closing) to remove speckles and fill holes.

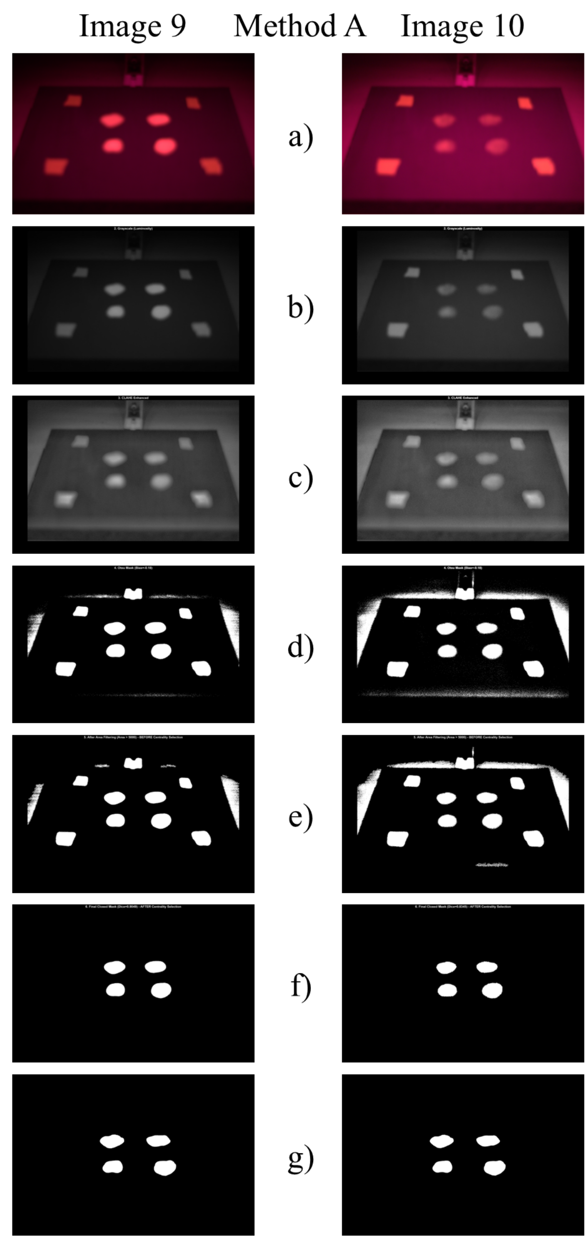


Figure 1. Method A pipeline as documented by Felipe: original UV-C image, contrast enhancement, biased thresholding, filtering, and the cleaned binary mask (panels a–g).

2.4.3. Discussion

Experiments with $K = 2, 3, 4, 5$ show that $K = 3$ is a good precision/recall balance, $K = 4$ retains high recall while sacrificing some precision, and higher values lose too much recall. Table 1 reports the metrics for the $K = 3$ case without pre-processing; the variants that applied filtering produced lower accuracy and recall.

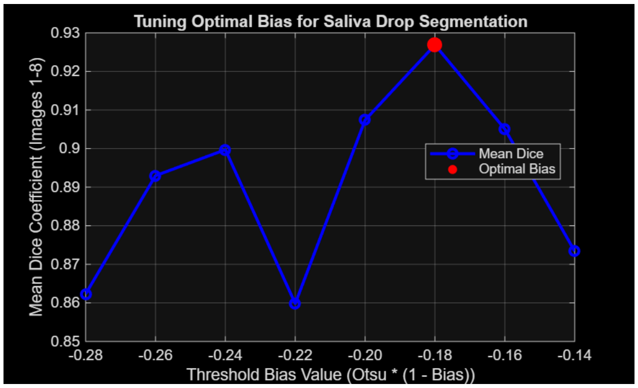


Figure 2. Bias parameter selection used to shift Otsu's threshold toward a more permissive mask (Figure 3).

Table 1. Method C ($K = 3$) metrics without pre-processing.

Image	Accuracy	Precision	Recall	Dice
1.tif	0.997	0.937	0.896	0.916
2.tif	0.844	0.092	0.820	0.165
3.tif	0.990	0.813	0.949	0.876
4.tif	0.960	0.432	0.895	0.583
5.tif	0.818	0.096	0.995	0.175
6.tif	0.779	0.105	0.985	0.190
7.tif	0.987	0.671	0.923	0.777
8.tif	0.949	0.387	0.977	0.554
9.tif	0.981	0.552	0.989	0.709
10.tif	0.963	0.358	0.820	0.498
Average	**0.927**	**0.444**	**0.925**	**0.544**

2.5. Method D: probabilistic Gaussian mixture model

Method D upgrades the k-means clusters with a full Gaussian mixture model that quantifies uncertainty and chooses the most probable segmentation. Each pixel $\mathbf{x}_i = (R_i, G_i, B_i)^\top$ is modeled as

$$p(\mathbf{x}_i \mid \Theta) = \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x}_i \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

with $K \in \{3, 4\}$, full covariances $\boldsymbol{\Sigma}_k$, and mixture weights π_k. Expectation–Maximization is handled by `sklearn.mixture.GaussianMixture`, while our contribution is the multi-start setup, initialization, and model selection via pseudo-log-likelihood.

2.5.1. Pre-processing

- The images remain in RGB space and are reshaped into pixel arrays so the GMM learns the native color distribution.
- Candidate means $\boldsymbol{\mu}_k^{(0)}$ come from Method C's k-means centers, helping EM start near saliva/background splits.
- No additional smoothing or filtering is applied, keeping preprocessing comparable to the other pipelines.

2.5.2. Training and model selection

- For each image and each K, we run $n_{\text{runs}} = 10$ independent GMM fits mixing k-means seeds with random starts to explore multiple basins of attraction.
- Each fit returns

$$\text{PLL}(\Theta) = \sum_{i=1}^{N} \log p(\mathbf{x}_i \mid \Theta),$$

  which we use to select the most probable solution because EM only guarantees local convergence.
- EM yields responsibilities

$$\gamma_{ik} = p(z_i = k \mid \mathbf{x}_i, \Theta^*),$$

  so the foreground probability map $P_{\text{FG}}(\mathbf{x}_i)$ is given by $\gamma_{ik_{\text{fg}}}$, where k_{fg} is the component with the highest luminance $Y_k = 0.299R_k + 0.587G_k + 0.114B_k$.

2.5.3. Mask construction and evaluation

- We threshold the probability map at 0.5, yielding a soft mask that is refined with morphological opening and closing to remove noise.
- Accuracy, precision, recall, Dice, and the pseudo-log-likelihood of the selected model are recorded for every image and every K; these per-image metrics feed the results summary and demonstrate how likelihood-guided selection aligns with segmentation quality.

This fully probabilistic workflow keeps the interpretability of k-means while adding soft assignments, likelihood-based selection, and quantified uncertainty via the posterior probability maps.

3. Results and discussion

Table 2 summarizes the average accuracy, precision, recall, Dice, and pseudo-log-likelihood (PLL) for the evaluated configurations. Method C delivers the highest precision at $K = 3$, while the GMM maintains recall above 0.96 even when precision drops; switching to $K = 4$ raises Dice and improves the PLL, signaling a tighter probabilistic fit to the RGB distributions.

All per-image accuracy, precision, recall, Dice, and PLL values are recorded for every run so that the aggregated numbers in Table 2 and the qualitative observations align with the logged findings. These per-image logs capture the variability across substrates and support reproducibility, allowing us to trace the numbers in the table back to the recorded output of each experiment.

Method	K	Accuracy	Precision	Recall	Dice	PLL (10^6)
Method C (k-means, no preprocessing)	3	0.927	0.444	0.925	0.544	–
Method D (GMM)	3	0.930	0.282	0.992	0.432	-2.92
Method D (GMM)	4	0.959	0.419	0.964	0.564	-2.77

Table 2. Cross-method comparison of segmentation metrics. The GMM keeps recall high and the $K = 4$ solution improves Dice while increasing PLL relative to $K = 3$.

The probabilistic GMM absorbs bright saliva pixels that the hard clusters sometimes treat as background, yielding near-perfect recall at $K = 3$. The addition of a fourth component, chosen by the highest PLL over ten EM runs (seeded by both k-means centers and random starts), regains precision and raises Dice while still delivering high recall. This likelihood-guided selection, reinforced by the probability maps and morphological cleanup, ensures that each reported mask aligns with the PLL value that drove its selection.

Figure 3 showcases representative outputs from Image 1 for $K = 3$ and $K = 4$; the probability maps illustrate how the brightest Gaussian concentrates on saliva, while the thresholded masks show how morphological cleanup delivers the final binary regions.

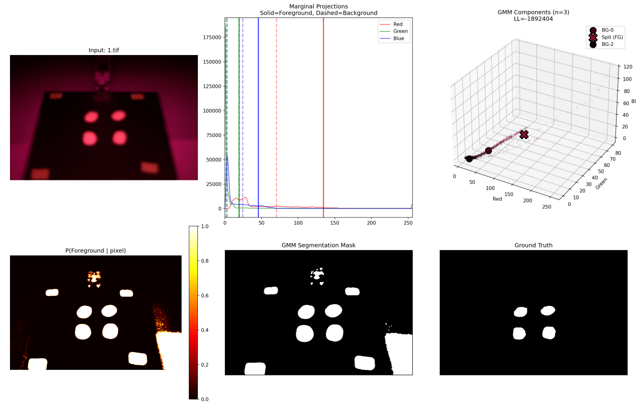
Qualitative outputs (probability maps and binary masks) complement the numeric metrics: the posterior probabilities provide spatial confidence and the thresholded masks, once cleaned with opening/closing, reflect both the high-likelihood components and the morphological heuristics.

4. Discussion

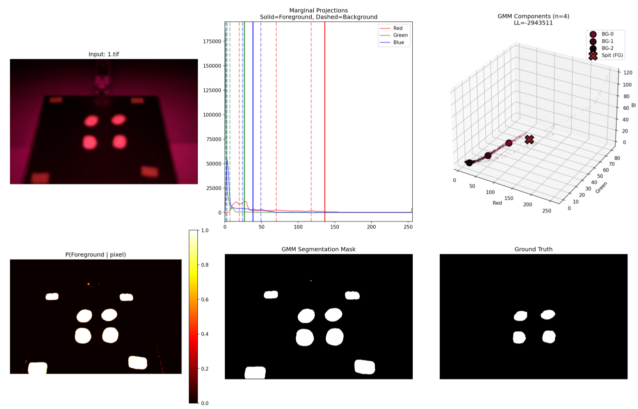
Method C offers a fast, hard-clustering baseline: $K = 3$ keeps precision around 0.44 while $K = 4$ shifts toward higher recall. Method D embraces the uncertainty of Gaussian mixtures, turning k-means centers into soft assignments while the PLL separates the best runs from the rest. The per-image metrics show that the probability map recovers gaps that hard clustering misses, keeping recall above 0.96 even as Dice rises at $K = 4$.

Recording accuracy, precision, recall, Dice, and PLL after each fit makes the pipeline auditable, so readers can reproduce the trade-offs and confirm that the chosen components correlate with better log-likelihoods. Pending runtime and memory profiling numbers will extend the comparison to computational cost, but for now the focus remains on stable segmentation in the low-data regime.

Future work will expand the dataset and explore hybrid models. Collecting more UV-C frames (or generating synthetic droplets) would help validate supervised backbones such as U-Net or transformer encoders, while additional substrates and illumination settings will test generalization. We also plan to investigate domain adaptation or attention-guided priors so the models can focus on saliva-like blobs despite new backgrounds, and to capture runtime/memory inferences on the target hardware so we can report deployment costs alongside accuracy and PLL.



(a) $K = 3$ probability map + mask



(b) $K = 4$ probability map + mask

Figure 3. Gaussian mixture outputs for Image 1 at $K = 3$ and $K = 4$. The probability maps highlight the saliva component, while the cleaned masks demonstrate the final segmentation.

5. Conclusion

This report documents an unsupervised saliva segmentation workflow that combines Method A's MATLAB thresholding, Method C's k-means clustering, and Method D's likelihood-aware Gaussian mixture model. Method D keeps recall high while raising Dice at $K = 4$ by selecting the most probable EM run via pseudo-log-likelihood, illustrating that probabilistic modeling and multi-start EM deliver stable masks for safety-critical contamination imagery. The detailed log of per-image metrics ensures the reported averages trace back to reproducible per-run outputs.

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